

next to machine life forms, nano-enhanced superhumans, virtual immortality, overwhelming material abundance, and interplanetary colonization. A world where these capabilities were real might, over time, become unrecognizable to people alive today lacking in all the familiar landmarks of everyday experience and common sense. The normal rules of social interaction wouldn't apply when humans share the solar system with non-human life forms of unspeakable power. Customs revolving around the normal cycles of birth, growth, maturity, and death would be obsolete when intelligent beings live for hundreds or thousands of years or longer. The traditional human process of working for a living would become irrelevant if material goods materialized on demand out of vast, living swarms of nanomachines. Arts, skills, and crafts would be rendered meaningless.

If nanotechnology manages to unleash technological changes incomparably more immense than any seen before, the word "human" might become an empty, obsolete expression. If history is any indication, we can't discount the possibility of violent convulsions as old institutions and values collapse. The end of feudalism, the birth of Protestantism, and the expansion of industrial capitalism all witnessed violent resistance from the defenders of an old order being swept away. The building of the first nanorobot, if it occurs, will usher in the same kind of transition period, during which the meaning and future of nanotechnology will hang in the balance. Even in its infancy, in fact, the field has already sparked opposition, with left wing groups calling nanotechnology a symbol of scientific arrogance and the harbinger of a techno-dystopia.

A more subtle threat is that represented by the cyborg. Taking nanodevices into our bodies, either to extend our powers or to extend our lives, creates an intimate relationship between our machines and us. Whether the result is viewed as symbiotic or parasitic, at some point, the needs, values, and orientation of these new individuals may become drastically different from what is currently defined as human. To some, this is an opportunity to revel in, but there is by no means a consensus as to whether this is good for our species or not. The social consequences could be profound:

- ▶ Are unmodified humans obsolete (and possibly expendable)?
- ▶ As people become cyborgs, are they still part of our community with all the same rights?
- ▶ How would the accumulation of power and wealth by "immortals" be handled?
- ▶ Who is responsible for the costs of side effects?
- ▶ Philosophically, if all natural parts are replaced by nanodevices, and the resulting individual passes the Turing test, do you still have a human?